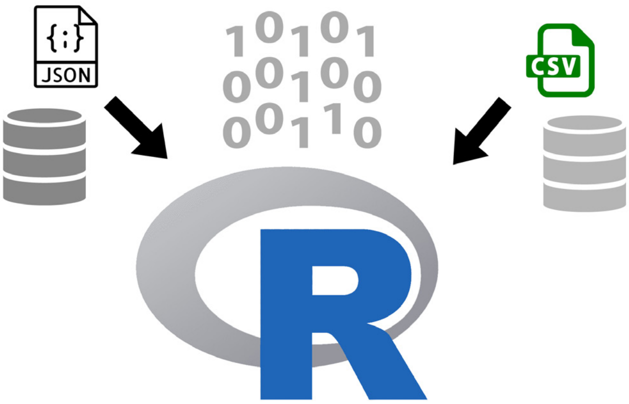
Importing Data in R

In this fourth article in the 'R, Statistics and Machine Learning' series, we shall explore the various ways to import data in R.



R provides various functions to load data. We shall explore reading data from CSV, JSON files and databases. We will be using R version 4.1.0 installed on Parabola GNU/Linux-libre (x86-64) for the code snippets.

```
$ R --version
R version 4.1.0 (2021-05-18) -- "Camp Pontanezen"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
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```

CSV

Consider the 'Bank Marketing Data Set' available from the UCI Machine Learning Repository at *https://archive.ics.uci.edu/ml/datasets/Bank+Marketing*. The data set is from a Portuguese banking institution and is available freely for public research use. There are four data sets available, and we will use the *read.csv()* function to import the data from a 'bank.csv' file into a data frame, as shown below:

```
> bank <- read.csv(file="bank.csv", sep=";")

> bank[1:3,]
  age         job marital education default balance housing
loan  contact day
1  30 unemployed married   primary      no    1787      no
no cellular  19
2  33  services married secondary      no    4789     yes
yes cellular  11
3  35 management  single  tertiary      no    1350     yes
no cellular  16
  month duration campaign pdays previous poutcome  y
1   oct       79        1    -1        0  unknown no
2   may      220        1   339        4  failure no
3   apr      185        1   330        1  failure no
```

There are multiple options that you can pass to the 'read.csv' method. They are explained below:

- *file:* This represents the source of the file name from which the data is to be read. In our example, it is 'bank.csv'.
- *header:* This is a Boolean condition which specifies if the file contains the names of the columns. In 'bank.csv', the header consists of:

```
> colnames(bank)
 [1] "age"       "job"       "marital"   "education"
"default"   "balance"
 [7] "housing"   "loan"      "contact"   "day"       "month"
"duration"
[13] "campaign"  "pdays"     "previous"  "poutcome"  "y"
```

- *sep:* This is the character that separates the various fields in the file. For example, comma (",") for CSV files.
- *quote:* This specifies the type of quotes to be used if the characters are enclosed within a quotation.
- *dec:* Indicates the character to be used to separate decimal points. In Europe, the practice is to use comma